

IPS : Integrating Pose with Speech for enhancement of body pose estimation in VR remote collaboration

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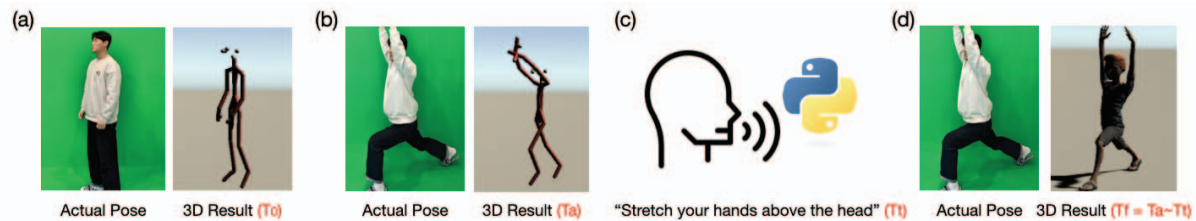


Figure 1: Overview of our system. (a) Normal pose estimation without any detection issues. (b) Capturing the last pose when the system detects camera out of sight. (c) Activation of speech recognition and target pose generation from motion library. (d) Pose smoothing between the last pose captured by camera and the target pose generated from speech.

ABSTRACT

We propose a Speech-Pose integration method to overcome the limitation of existing body pose estimation. Unlike previous Speech-based gesture generation method, our proposal reflects the user's actual pose using a vision-based system and speech as a subsidiary. When the system detects the camera out of sight, we carry out a context-aware method analyzing the speech. The system determines the target pose and connects it with the last pose captured by the camera using the bounding box. Our system can be a robust solution for avatar-mediated remote collaboration which requires accurate gesture delivery such as VR remote yoga training.

1 INTRODUCTION

As Mixed Reality (MR) technologies have improved a lot in the past decade, there are various approaches to providing remote collaboration by representing the users as avatars. While there are many remaining issues in terms of representing avatar non-verbal communication cues such as body gestures and facial expressions, we focus on body gesture, which is an essential part of providing meaningful and unique information to a spoken message [2]. In the human pose estimation method capturing the body movement and representing users' gestures, out-of-camera sight is a continuing problem [9]. Avatar poses with lower estimation accuracy cannot represent what users truly meant to express, often bringing misunderstanding or misinterpretation. Therefore, it can be a critical issue interrupting remote collaboration in immersive environments.

As speech shows a high correlation with gesture [4, 5, 8], previous work proposed a variety of speech-to-gesture methods. While several works [1, 3] proposed speech-based gesture generation methods, they commonly do not consider real-time video input, not providing actual human poses but generating gestures from speech only. Though Rebol et al. [6, 7] proposed a 3D gesture prediction system that synthesized the 3D human pose estimator and GAN based model, they are limited to generating beat gestures, dealing with gesturing styles depending on the voice and prosody. To be applied

to human to human remote collaboration representing actual human poses with an avatar, we propose a gesture integration method using speech input as a subsidiary for gesture generation, pointing out semantically important information from a communication context.

2 PROPOSED SYSTEM

A problem we focused in human pose estimation using a single RGBD camera is a limited field of view, which can fail to capture body parts that are out of sight or above the head [9]. When our body parts go above the camera, it simply cannot detect our pose and the avatar gestures can be represented in a strange form. It can be critical when delivering semantic information through accurate gesture communication. Speech being a robust solution for the camera out-of-sight issue, we propose a speech-gesture integration method using speech as a subsidiary gesture generation.

The system primarily generates poses with Azure Kinect and trackable poses are estimated as a 3D result. However, if the system detects out-of-camera sight with the implemented bounding box, the system activates a speech-to-pose mode. Extracted audio sources are converted into text understanding what it means. Rather than transferring whole speech data to gesture, we selectively use speech when it detects challenges. For the speech-to-pose method, we propose a context-aware method, which enables setting the target pose with analysis of the semantic information. One example could be "Stretch your hand above the head". By analyzing each component in the sentence, we could see that verb and direct object decide the specific movement or motion, and preposition and complement decide the direction and exact target position. So we can finally decide the target pose from the context-aware method. Connecting the target pose from the last pose captured by the camera pose right before the issue detection, we use pose smoothing to show the gesture in a stable and accurate form. Our approach enables pose generation from the main use of actual human poses, and therefore the system is more suitable for human-to-human interaction than other speech gesture generation methods.

3 POTENTIAL USER SCENARIO

Our system has the potential to overcome the challenge of camera out-of-sight issues. While capturing one's pose with one single camera in a limited space often brings some limitations, we decided to focus on the out-of-camera sight issue. Therefore, we found a potential user scenario as VR yoga training, which requires accurate

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and delicate avatar pose generation. In VR remote yoga training, the trainer needs to show a demonstration to the trainee. The upper body part or the back of the user cannot be tracked and therefore the avatar gestures cannot be accurately represented. Since it requires an accurate pose estimation in the case of avatar gesture generation, we think our proposed system is suitable due to its robustness. Representing the trainer as an avatar from the remote space to the local space, we enable the trainer to help a trainee's pose with accurate pose generation.

4 PROTOTYPE

Trackable poses are estimated as a 3D result by the existing exocentric body pose estimation method with Azure Kinect. If the system detects out-of-camera sight and the direction where the user's body part went through, it immediately triggers a speech-recognition library for audio source conversion into text data. Using the direction of collided box and text input, the target pose will be generated from the motion capture pose dataset. The following concepts are used for the detailed implementation.

- T_0 is an initial user pose with 3D result from exocentric pose estimation.
- T_a is an estimated pose that got out of camera sight.
- T_t is a speech-generated target pose from the speech recognition library.
- T_f is a target pose generated after pose smoothing.

The overall system diagram is described in figure 2. We use the Azure Kinect RGB-depth sensor as our main camera to estimate full body pose from an exocentric view. A single RGBD camera is placed 1 meter away from the user. With this setup, the user's risen hands are been cut off due to Azure Kinect's FOV limitation allowing us to observe out-of-sight issues. As shown in figure 1 of (b) Out of camera sight, the estimated 3D body pose result on the right is not estimated correctly from the actual body pose on the left.

And because of this limitation, we use the Unity game engine to create an invisible bounding box in front of the users' chest position, which is approximately 140cm above the floor. Each box is located at the up, down, right, and left sides of the user, which detects out-of-camera sight, representing the edge of Kinect's limited view. So, when the user moves their hands from the bounding box to its borderline, the box will notify exact direction information.

Speech-to-Text conversion is done by the SpeechRecognition python3 library. The library uses Google Web Speech API to convert Audio data into text data and recognizes predetermined specific keywords. The converted text is communicated to the Unity application for further processing. And the yoga motion capture data is used for target pose generation. Specific keywords, in this case, "above", "up" and "raise", are connected to contextually appropriate poses. The final pose is then updated.

5 CONCLUSION AND FUTURE WORK

We proposed a Speech-Pose integration method using speech as a subsidiary method to get over the out-of-camera sight issue of existing body pose estimation methods. When out-of-camera sight is detected, the system triggers speech recognition and set the target pose from the motion library. Then the system smooth the poses, connecting the last pose from the camera view to the target pose. Using a context-aware method of analyzing speech, the system can be a robust solution to various vision based issues, which can be critical especially in situations where dynamic user movement occurs in a limited space.

To leap forward, we should address some limitations and future works. In terms of the system, it has limitations since the avatar

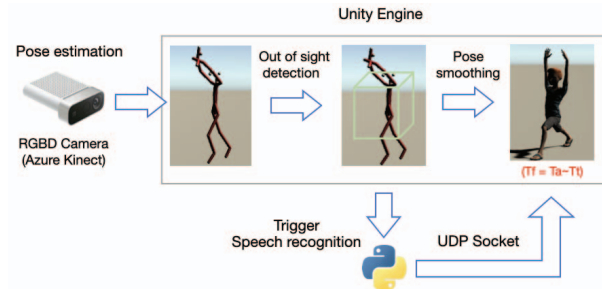


Figure 2: System Overview

can only present predefined motion. Currently, the only way to animate the user's action is to connect specific text to the box collider direction and animation. The speech recognition itself can detect numerous words which are pre-trained. However, our current system will not perform as intended if we do not set the specific words in the system before we run it. Also, since our prototype is currently getting speech recognition through Python-Unity communication, it has some delay in getting data from the python library. Therefore as future work, we should keep researching pose generation system that understands the users' intent and context. And connect it with deep learning methods, which can eventually understand full human body language without any additional input. By doing so, we are looking forward that the pose integration procedure can be reduced and being able to perform better in real-time.

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